Alex Bergman

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http://alex-lukas-bergman.github.io/

Current Occupation

2021 – · · · **PhD student** Lund university. My advisor is Professor Alexandru Aleman.

Research Publications

Preprints

1. A. Aleman and A. Bergman, *Invariant subspaces for generalized differentiation and Volterra operators*, 2025. arXiv: 2503.08443 [math.FA]. URL: https://arxiv.org/abs/2503.08443.
2. A. Bergman, *A remark on the Beurling-Malliavin theorem in several variables*, 2025. arXiv: 2512.07271 [math.CV]. URL: https://arxiv.org/abs/2512.07271.
3. A. Bergman, *Inner approximations of doubling weights with applications to Beurling-Malliavin theory in Toeplitz kernels*, 2025. arXiv: 2509.21229 [math.CA]. URL: https://arxiv.org/abs/2509.21229.

Journal Articles

1. A. Bergman, "Hörmander's inequality and point evaluations in de Branges space," *Rev. Mat. Iberoam.*, 2025, published online first. DOI: https://doi.org/10.4171/rmi/1582.
2. A. Bergman, "A sharp entropy condition for the density of angular derivatives," *C. R. Math. Acad. Sci. Paris*, vol. 363, pp. 29–33, 2025, ISSN: 1631-073X,1778-3569. DOI: 10.5802/crmath.695.
3. A. Bergman, "On cyclicity in de Branges-Rovnyak spaces," *Indiana Univ. Math. J.*, vol. 73, no. 4, pp. 1307–1329, 2024, ISSN: 0022-2518,1943-5258. DOI: 10.1512/iumj.2024.73.60011.
4. A. Bergman and O. Rubin, "Chebyshev polynomials corresponding to a vanishing weight," *J. Approx. Theory*, vol. 301, Paper No. 106048, 25, 2024, ISSN: 0021-9045,1096-0430. DOI: 10.1016/j.jat.2024.106048.

Academic Talks

2025

- **Hörmander's inequality in de Branges space**, Analysis seminar, Bilkent university
- **Hörmander's inequality in de Branges space**, Analysis seminar, Lund university
- **Hörmander's inequality in de Branges space**, Conference "Hilbert Function Spaces", Rome, Italy.
- **Hörmander's inequality in de Branges space**, Conference "Math bench sessions", Krakow, Poland.

2024

- **Hörmander's inequality in de Branges space**, Analysis seminar, KU Leuven, Belgium.

Academic Talks (continued)

- **A sharp entropy condition for the density of angular derivatives**, Conference "Operators and analytic function spaces", Luminy, France.
- **Smooth invariant chains in de Branges space**, Conference "Advanced courses in operator theory and complex analysis", Tenerife, Spain.
- **Invariant subspaces for generalized differentiation and Volterra operators**, Analysis seminar, NTNU, Norway.
- **Invariant subspaces for generalized differentiation and Volterra operators**, Analysis seminar, King's college London, England.
- **Cyclic Vectors in de Branges-Rovnyak Spaces**, Analysis seminar, Lund university, Sweden.

2023
- **Smooth invariant chains in de Branges space**, Conference "Math bench sessions", Krakow, Poland.
- **Invariant subspaces for second order ODE**, Analysis seminar, Stockholm university, Sweden.
- **Invariant subspaces for second order ODE**, Conference "International Conference on Spectral Theory and Approximation", Lund, Sweden.
- **On cyclicity in de Branges-Rovnyak spaces**, Conference "Advanced courses in operator theory and complex analysis", Thessaloniki, Greece.

2022
- **Interpolation spaces between H^1 and H^∞**, Math seminar, Umeå university, Sweden.

Grants

2024
- **The Fund of the Torsten and Fanny Brodén Foundation** General research grant.
  The Royal Physiographic Society of Lund.
- **The Bokelund Society Travel Grant** Travel Grant.
  Faculty of Science, Lund university.

2023
- **The Fund of the Walter Gyllenberg Foundation** General research grant.
  The Royal Physiographic Society of Lund.

2022-2024
- **Thorild Dahlgrens och Folke Lannérs stipendiefond** Travel grant.
  Awarded several times.

Academic Visits

Research Stays

2024
- **Research stay hosted by Profesor Alexei Poltoratski**, November 2024, University of Wisconsin, Wisconsin, USA.

Academic Visits (continued)

- **Research stay hosted by Professor Alexander Pushnitski**, May 2024, King's college, London, England

Summer Schools

2023 - **Analysis, arithmetics and geometry: zeta-functions, trace formulas, and around**, Mittag-Leffler institute, Stockholm, Sweden.

2022 - **Summer school in Harmonic analysis**, Bonn/Kopp, Germany.

Teaching

2025 - **Linear algebra** Fall, Lund university.

- **Linear analysis** Fall, Lund university.

- **Envariabelanalys (analysis in one variable)** Spring, Lund university.

2024 - **Analysis in several variables I** Fall, Lund university.

- **Analysis in one variable** Fall, Lund university.

- **Envariabelanalys (analysis in one variable)** Spring, Lund university.

2023 - **Analysis in one variable** Fall, Lund university.

- **Envariabelanalys (analysis in one variable)** Spring, Lund university.

2022 - **Analysis in several variables II** Fall, Lund university.

- **Analysis in one variable** Fall, Lund university.

- **Envariabelanalys (analysis in one variable)** Spring, Lund university.

2021 - **Analysis in one variable** Fall, Lund university.

Referee Work

I have been a referee for the following journals.

Analysis and Mathematical Physics

Education

2018-2021 From 2018 to 2021 I was a student at the department of mathematics at Lund university.

Master thesis (2021): *Interpolation spaces between Hilbert and Banach spaces of analytic functions*
Bachelor thesis (2020): *Weak type spectral mapping theorems for strongly continuous groups of operators*

Community outreach and Popular Science

2024 **Forskaruppsats** I participated in an interview study for high school students at "Polhemskolan" (Lund, Sweden) aspiring to be scientists.

Question and answer session. I answered questions from high school students in Skåne, Sweden about mathematics over the phone.

2023 **Question and answer session.** I answered questions from high school students in Skåne, Sweden about mathematics over the phone.

Other departmental duties

I was a member of the following hiring committees

2025 PhD student in dynamical systems at Lund university.

2023 PhD student in representation theory at Lund university.

I am or have been a member of the following boards.

2024-··· I am currently a member of the Mathematics program council at Lund university. The purpose of the council is education development. My role is as the PhD student representative.